

Yagyaansh Khaneja

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Education

IIT Roorkee, B. Tech. in Engineering Physics (CGPA 7.11/10)	2021
RDPS, Delhi (CBSE), Class XII: PCM (94.2 %)	2017
RDPS, Delhi (CBSE), Class X (CGPA 10/10)	2015

Skills

- Programming:** Python (PyTorch, NumPy, scikit-learn, pandas, matplotlib), FORTRAN, HTML/CSS, SQLite
- Tools:** MATLAB, COMSOL, CST MW Studio, Excel/Word/PowerPoint, LaTeX, Lightroom
- Languages:** English (Native), Hindi (Native), French (Basic), Punjabi (Basic)

Certificates

- Machine Learning:** Andrew Ng ([Coursera](#)) [2023]
- Python for Everybody:** Charles Severance ([Coursera](#) Specialization) [2020]
- Web Design for Everybody:** Colleen ven Lent ([Coursera](#) Specialization) [2020]
- Internet: History, Technology, Security:** Charles Severance ([Coursera](#)) [2020]

Achievements

- NGPE 2019 (Part A):** Top 10%
- JEE 2017 (Advanced):** AIR 3188
- JEE 2017 (Mains):** AIR 2933 (Top **0.2%**)
- Vijyoshi 2016 (Camp):** Invited Guest
- KYPY 2015 (SA):** AIR 300 (Top **0.4%**)
- NTSE 2015 (II):** Scholar (Top **0.1%**)

Independent Study

- CS50 (Stanford):** [2020]
- CS50 (Stanford):** AI with Python [2020, 2021]
- Andrej Karpathy:** NN: Zero to Hero [2024-25]
- 3Blue1Brown:** DL & Transformers [2025]
- 6.S191 (MIT):** Intro to Deep Learning [2025]
- Kaggle:** Kaggle Learn [2025]

Tech Projects

- **Vibe Coding Engineer** [Oct 2025 – present]
 - **Organization:** US based stealth StartUp (San Fransisco)
 - **TL;DR:** Built automation tools for Hotel industry
 - Developed playwright scripts (Python) to scrape data from Property Management Systems (PMS) like ChoiceAdvantage, IHG HolidayInn, etc.
 - Built a robust **reconciliation** engine to find discrepancies in data scraped from various HMS, CLC, Expedia, Booking.com, etc.
 - **Results:** Got first 2 customers signed up within 1 week of prototype devt.
 - **Skills:** AWS Deployment, Claude Code CLI, Cursor/Windsurf/VS Code, OpenAI Codex, Processing webpages, data (json, csv) in Python
- **Project Manager (Volunteer) – Tech Team** [Dec 2024 – Jun 2025]
 - **Organizations involved:** [Tarayana](#), [GIAOS](#), [Omniverse](#)
 - **TL;DR:** Guided the development of a [data dashboard](#) (PoC) for a remote village in Bhutan
 - Empowered Rukha Village (Wangdi), Bhutan in leveraging technology to advance the national vision of Gross National Happiness (GNH)
 - Under the aegis of renowned banker Tan Sri [Andrew Sheng](#) SBS
 - **Skills:** Team Management, High Level Architecture Design, (Tech Stack: React Native, MongoDB, DeepSeek API)

Multidisciplinary Projects

- **Family Business – B2B Wholesale (pan India)** [Sep 2022 – present]
 - **Location:** Asia's largest wholesale market: Chandni Chowk/Chawri Bazaar (Delhi-6)
 - **TL;DR:** In essence, increased the efficiency of operations, **saving 4 hours per day**
 - Migrated **accounts** from legacy (handwritten) to cloud software, enabling remote work for the first time in 35 years.
 - Meticulously classified 1000s of SKUs (**Steel products**), 400+ parties; making data entry organized
 - **Reverse engineered** API for a third-party accounting app for reconciliation!
 - Cut operational overhead via warehouse optimization (Streamlined packaging, labels, item codes, colour coded dispatch, and process flows)
 - Expanded customer base through **online retail** sales via [Meesho](#)
 - Designed a [catalog](#) of Door Fittings, Fasteners, Screws - simplifying products for customers (Canva)
 - Learned invaluable lessons about Money, **Cash Flow**, Relationship Management, Business Ethics, Sales, etc.
 - **Results:** Revenue scaled to \$1M USD within a year of joining
 - **Skills:** Warehouse Management, Product Design, Accounting, Financial Management, GST, e-invoice, e-way bills
- **Research Intern (Energy Team) – Sonam Wangchuk** [Jan 2022 – May 2022]
 - **Organization:** Himalayan Institute of Alternatives ([HIAL](#)), Ladakh
 - **TL;DR:** Assisted renowned environmentalist [Sonam Wangchuk](#) on his projects (SDGs)
 - Drafted [manual](#) for construction of Passive Solar Heated ([PSH](#)) Buildings in Ladakh.
 - Energy Analysis of PCM in Tromb  Walls (PSH), and HVAC systems
 - Maintained and installed solar PV systems (20 kW, 7.5 kW)
 - Monitoring & Data Processing for Lead-Acid battery house (240V DC to AC).
 - Installed new Stevenson Screen; assisted in weather [data collection](#) (& analysis) in remote regions of Ladakh using SensorPush and Campbell Scientific weather stations.
 - **Skills:** Research (Thermal Physics), Excel, Project Implementation, Electrical Science

Side Projects

- **Tracking Pixel** [one fine night in Sep 2025]
- **TL;DR:** Made a tracker.js file for use in emails/website
 - Made use of a transparent .png pixel to log GET requests
 - Used Firebase function (JS) to listen & log requests
 - Stored the logs in Firestore collection for querying/analysis
- **Skills:** Firebase Hosting, JavaScript, Firestore DB

- **Standard Model – Visualized** [May 2025 - ongoing]
- **TL;DR:** Visualized Standard Model of Particle Physics in 3D
 - Used three properties at a time (mass, charge, isospin/strangeness)
 - Tried matplotlib [initially](#), then switched to Manim for better [animations](#)
- **Skills:** Python – matplotlib, Manim (by 3Blue1Brown)

- **Personal Website – yagvaansh.com** [Nov 2020 – Jan 2021]
- **TL;DR:** Hard-coded my website using basic HTML, CSS and JavaScript (no frameworks)
 - Implemented Dark Theme toggle using JavaScript
 - Wrote blogs with graphics, integrated comments section using Disqus
 - Hosted my website using GitHub Pages
- **Skills:** Web Development (HTML, CSS), handling SVG in HTML

- **Fee Data Extraction** [Aug 2020]
- **TL;DR:** Made API calls (Azure server) to extract data of 3000+ students (college fee receipts)
 - Libraries used - urllib.request, ssl, json, sqlite3 in Python
 - converted the received json data into a SQLite database for easy accessibility
- **Skills:** Handling JSON and SQLite database using Python

Physics Projects

- **Bachelor's Project – Silicon Photonic Switch, IIT Roorkee** [Dec 2018 – Sep 2021]
- **TL;DR:** Simulated non-volatile GST-SOI waveguides using CST MW Studio and COMSOL.
 - Designed high performance photonic switches for use at $2.1\mu\text{m}$ ([report](#)) & 1 THz ([thesis](#))
 - Studied Mux-DeMux using Optical Phased Array & WDM for telecom applications
- **Skills:** EM Simulation ([COMSOL](#), CST MW Studio), MATLAB, Waveguide Photonics

- **Research Camp – Optics/Photonics, ITMO University** [May 2019 – Jun 2019]
- **Organization:** ITMO University, St. Petersburg, Russia
- **TL;DR:** Gained exposure to advances in Optics through [active engagement](#) with frontier research
 - Generated THz pulse radiation using laser filamentation of liquids. ([First global attempt](#) using pH variation of water!)
 - Simulated interference & diffraction patterns for varying waveforms & apertures
 - Studied Quantum Information, Holography, QKD Systems, Femtosecond Lasers, etc.
- **Skills:** Experimental Optics, MATLAB, Laser Systems

Community Service

- **STEM & English Teacher** (May-Aug, 2022)
- **Locations:** Ladakh ([SECMOL](#)), Vietnam ([Workaway](#))
 - Taught dropouts using hands-on science/math [experiments](#).
 - Led astronomy and photography hobby clubs.
- **Skills:** Communication, Empathy

- **Leadership Roles, IIT Roorkee** (2017-2021)
 - Additional Secretary, Photography Section
 - Coordinator, Thomso (Cultural Fest)
 - Content Manager, Srishti (Tech Exhibition)
- **Skills:** Leadership, Management